(2.)

$$W_P(s) = \frac{3s + k_1}{s^3 + 3s^2 + (2k_2 - k_1)s + k_2}$$

$$f(s) = 3s + k_1 + s^3 + 3s^2 + (2k_2 - k_1)s + k_2$$

$$f(s) = s^3 + 3s^2 + (2k_2 - k_1 + 3)s + k_2 + k_1$$

Routh

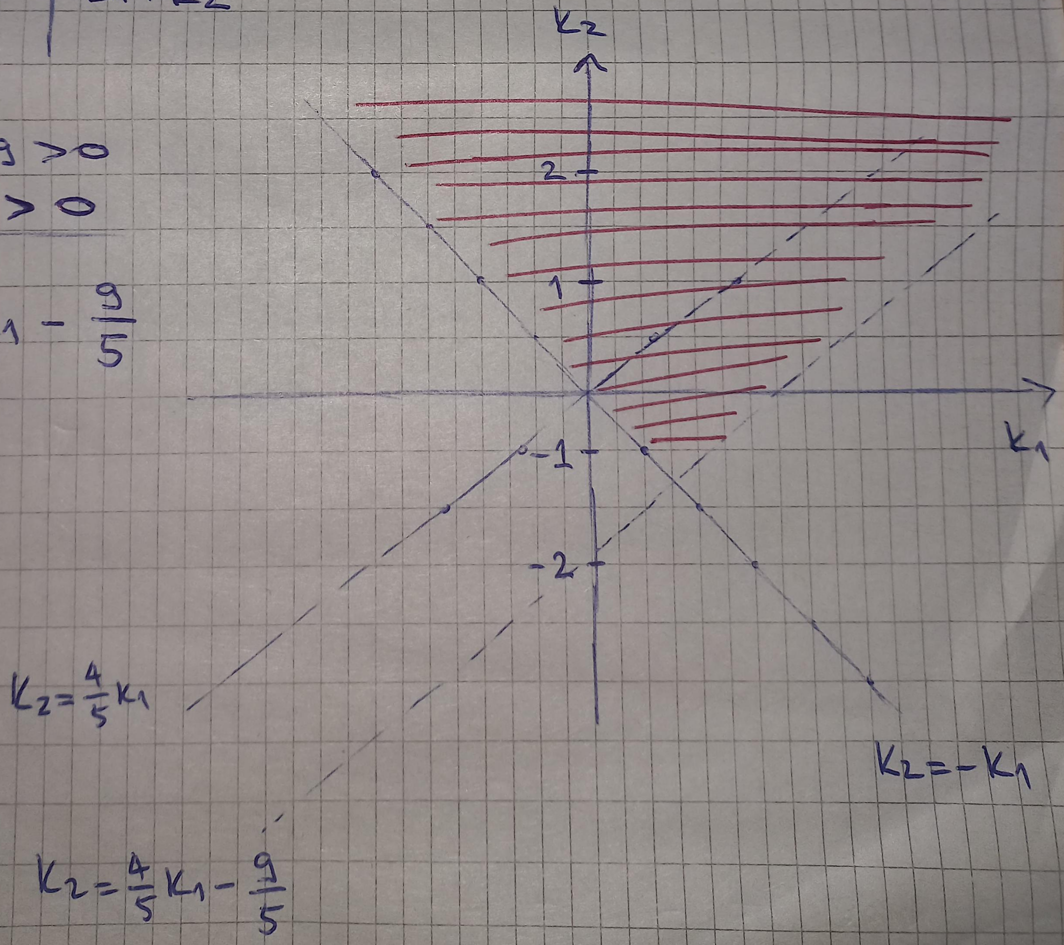
s^3	1	$2k_2 - k_1 + 3$
s^2	3	$k_1 + k_2$
s^1	$\frac{5k_2 - 4k_1 + 9}{3}$	
s^0	$k_1 + k_2$	

$5k_2 - 4k_1 + 9 > 0$

$k_1 + k_2 > 0$

$$k_2 > \frac{4}{5}k_1 - \frac{9}{5}$$

$$k_2 > -k_1$$



(1.)

$$W_S(s) = \frac{2s+3}{s^3+4s^2+(K_1+K_2)s+2K_1+3K_2}$$

$$f(s) = s^3+4s^2+(K_1+K_2)s+2K_1+3K_2$$

Routh:

$$\begin{array}{c|cc} s^3 & 1 & K_1+K_2 \\ s^2 & 4 & 2K_1+3K_2 \\ s^1 & \frac{2K_1+K_2}{4} & \\ s^0 & 2K_1+3K_2 & \end{array}$$

$$2K_1+K_2>0$$
$$2K_1+3K_2>0$$

$$K_2 > -2K_1$$
$$K_2 > -\frac{2}{3}K_1$$